



Fig. 4: RFID based access control device



Fig. 5: Electromagnetic lock installation on a glass door

open the door, the lock should open and then close automatically after a few seconds to avoid tailgating. For this, we need the relay circuit.

Most of the biometric devices or access control readers are pre-equipped with a relay circuit on board. The circuit is designed in such a way that the GND rail from the power supply/adaptor connects through the NC and COM terminals of the reader and exit push button. The +12V DC from the adaptor or SMPS is connected directly to the electromagnetic lock as shown in the block diagram (refer Fig. 1). Most of the magnetic locks and access control readers are pre-equipped with timer for locking and unlocking time delays. Generally,

presets are found on the PCB board, which can be adjusted for locking or unlocking time delays.

Working of access control system

When a pre-registered user or authorised person tries to open the door using an RFID tag (card) or password or biometrics, the relay circuit gets activated and the COM terminal gets disconnected from the NC terminal for a while. That is, the COM terminal gets connected to NO terminal. Hence, the 12V DC power supply is disconnected from the magnetic lock for a few seconds (depending on the preset time delay). So the magnetic lock gets unlocked when the power goes off. This action allows the door to open.

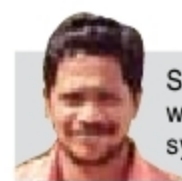
Again, when the user presses the exit push button, the relay gets activated and is forced to connect COM with NO terminal. The supply gets disconnected from the magnetic lock and it gets unlocked, the door gets opened again.

Basically, the access control device is to be installed near the door outside the room and exit button is to be installed near the door but inside the room. The electromagnetic lock installed on the glass door is shown in Fig. 5.

Some exit push buttons will have inbuilt relay circuitry, which must be given external power supply. In that case, it is preferred to use a 12V, 3A power supply.

RFID card. Some details on the RFID card/tag are given here. Every RFID card/tag has a unique number. When the tag is swiped across the RFID reader, the access control system prompts you to enter a password. The default password is 123456. To change the password, press #123456#0987654#987654# on the access control keypad. If you hear a long beep, it means the password is successfully registered. To add a new RFID card, press #987654#1 on the input keypad. Swipe the tag/card near the reader followed by pressing the # key.

EFY note. For more details on configuration and settings, refer RFID manual of this product at source.efymag.com. **EFY**



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